

Global frame selection and nonlinear link vacua

A permitted soft selector, a global quartic theorem and exact off-diagonal orbit certificates

PerfectPower research / 8 October 2026 (UTC)

The isolated link action admits a global unitary tensor-frame selection theorem. A complete certified finite-orbit census and massive-field relaxation expose the scale of its competing vacuum structure.

A permitted determinant A-term reverses the previously unfavorable common-mass ordering. A separate positive quartic completion gives an exact global frame-selection theorem for the isolated canonical link sector. Exact finite-group stabilizers enlarge the certified unperturbed census from 109 diagonal solutions to 130,681 matrices including the origin. The massless branch survives holomorphic massive-field relaxation through sixth order in the effective superpotential; its first possible effective F-term energy occurs at degree twelve. These results use the original sextic tensor and insert no CKM angle or golden coefficient.

Action, fields and independent operators

The link K is a complex three-by-three matrix with canonical Kahler metric. All source, Higgs and matter backgrounds vanish. The retained action is

$$W = a \det K + b \bar{l}_6(K) + c (\det K)^2, \quad V_F = \sum_{ij} |W_{K_{ij}}|^2.$$

The invariant has conjugated coefficients and holomorphic fields. Couplings a, b, c are real in the checked product-family CP convention. The existing exact Molien census gives one balanced quadratic, one holomorphic cubic and two balanced quartics. Consequently the complete nonconstant CP-even renormalizable scalar basis for this single link is N , $\text{Re } \det K$, N squared and $\text{Tr}(A \text{ squared})$, where $A = K^\dagger K$ and $N = \text{Tr } A$. Imaginary determinant couplings are excluded by the imposed CP symmetry. Quartic independence does not impose a relation between their coefficients.

Scalar mass and determinant A-terms are conventional soft supersymmetry-breaking interactions. General scalar quartics are not automatically soft supersymmetry-breaking operators. The positive completion below is stated as a scalar EFT interaction. Its Gram quartic equals a sum of formal $\text{SU}(3)$ moment-map squares, but continuously gauging $\text{SU}(3)$ while retaining a fixed non-invariant sextic tensor is not a consistent ultraviolet completion. A dynamical tensor or other gauge completion would have to address that separately.

An allowed soft term reverses the ordering

At $a=b=1, c=0$, denote the squared Frobenius norms of the equal-square group, distinct-square unitary and two-equal-square nonunitary branches by N_g , N_d and N_n . Their cubes are $27/1024$, $27/25$ and $(145+46 \sqrt{10})/160$. Their determinant real parts are $-1/32$, $-1/5$ and $-11/64$. These invariants are constant on each complete product-group orbit, including its off-diagonal members.

Consider the genuinely soft CP-even interaction

$$\Delta V = \varepsilon [-N - t \text{Re } \det K], \quad \varepsilon > 0.$$

The group orbit lies below both rival branches and the origin at first order whenever

$$\max\left(\frac{N_d - N_g}{1/5 - 1/32}, \frac{N_n - N_g}{11/64 - 1/32}\right) < t < 32N_g.$$

This is an open interval, approximately $6.5583 < t < 9.5244$. At $t=8$, first-order energies divided by epsilon are -0.0476377 for the group orbit, $+0.5740144$ for the distinct-square branch and $+0.1550993$ for the nonunitary branch; the origin has zero first-order energy. The receipt certifies the signs using rational bounds on every cube root and on $\sqrt{10}$. Thus a permitted determinant A-term can correct the wrong preference of a common tachyonic scalar mass alone. The coefficient is independent and allowed, rather than fixed by the symmetry.

This comparison applies to all four certified finite-group orbits. It is not a theorem that the soft-only action has no further lower vacua. In particular the massless branch cannot be assigned an ordinary nondegenerate perturbation expansion, and the origin can move under a tachyonic perturbation. The nondegenerate group branch persists for sufficiently small epsilon. The global theorem below supplies a stronger result for a specified completion.

A global quartic selection theorem

Assume a is nonzero, b c is positive, and λ and η are positive. Set $D_0 = -a/(32b+2c)$, and $v^3 = |D_0|$. Define the ordinary balanced quartic

$$G = \text{Tr} \left(A - \frac{N}{3} I \right)^2 \geq 0.$$

It vanishes exactly when K is a scalar times a unitary matrix. The completed scalar potential is

$$V_{\text{sel}} = V_F + \lambda(N - 3v^2)^2 + \eta G.$$

The shifted norm square includes a calibrated tachyonic mass, an independent positive norm quartic and an irrelevant constant. Its mass coefficient is $-6\lambda v^2$. Matching this radial scale to the F-flat group radius is a declared coupling relation, not an independently derived ultraviolet prediction. No orientation or CKM phase is chosen by that calibration. λ and η may vary independently throughout the positive quadrant without changing the zero-energy frame class.

Theorem. Every global minimum of V_{sel} has the form $K = r U$, where $r^3 = D_0$ and U is a determinant-one unitary stabilizer of the original conjugate sextic tensor. Conversely every such matrix has $V_{\text{sel}} = 0$. This zero set is finite and includes all 1,080 matrices in the checked conjugate 3.A6 orbit. Every nonunitary matrix, every origin solution and the previously identified distinct-square unitary branch is excluded from the global zero set. The theorem characterizes every global minimum; identifying the full stabilizer order with exactly 1,080 requires an additional stabilizer classification.

Proof. The three terms are nonnegative. The known group matrices achieve zero, so every global minimum has $V_F = 0$, $N = 3v^2$ and $G = 0$. Thus $K = v Q$ with Q unitary and $|\det K| = v^3$. Unitarity preserves the sextic tensor norm, whose square is sixteen. Cauchy-Schwarz therefore gives

$$|\bar{I}_6(K)| \leq 16v^6.$$

Euler's homogeneous identity at F flatness gives $D + 2b J + 2c D^2 = 0$, with $D = \det K$ and $J = \bar{I}_6(K)$. Write $D = D_0 \exp(i\delta)$ and $t = c/b$. Substitution gives

$$\frac{|J|^2}{v^{12}} = 256 + 2t(16 + t)(1 - \cos \delta).$$

Since $t > 0$ and the left side cannot exceed 256, δ is zero modulo a full turn. Hence $D = D_0$ and $J = 16 D_0^2$. Equality in Cauchy-Schwarz with this positive inner product forces the transformed conjugate tensor to be D_0^2 times the original tensor. Dividing K by any r with $r^3 = D_0$ yields U in $SU(3)$ preserving that tensor. Conversely tensor invariance differentiates to $\text{gradient } \bar{I}_6(r U) = 32 r$ to the sixth power times K inverse-transpose; the determinant gradient then makes all nine F equations vanish at the stated radius. The norm and Gram terms also vanish.

To prove finiteness without assuming the stabilizer order, differentiate the tensor action in eight real SU(3) tangent directions. Exact contraction gives

$$\|d\bar{T}(B)\|^2 = 72\|B\|_F^2$$

for every traceless anti-Hermitian B. All 64 Gram entries are verified in the exact coefficient field. The stabilizer has zero Lie algebra; as a closed subgroup of compact SU(3), it is finite. The potential is coercive because its positive norm square grows at infinity. Thus this is a global statement over all complex matrices, not a diagonal-ansatz comparison.

At $a=b=c=1$, $v^3=1/34$. The full original F terms vanish on the selected group orbit, while the known unitary and nonunitary competitors have positive completed energy. Direct product-group and generalized CP tests confirm that the completion retains the imposed symmetries. At a selected group minimum the original canonical masses are already strictly positive, and the added nonnegative terms cannot destabilize it.

Exact off-diagonal orbit census

The canonical action transforms K by $hL K hR^\dagger$, with hL and hR in the conjugate order-1,080 group. The determinant is invariant and the canonical field metric is preserved. For a nonsingular seed P, the stabilizer order is the number of group elements h for which $P h P^{-1}$ belongs to the group. This replaces a million-matrix duplicate search with 1,080 exact membership checks.

For the group seed the stabilizer order is 1,080, giving orbit size 1,080. For a two-equal-square nonunitary seed $\text{diag}(1,1,p)$, unitarity forces h to preserve the unequal singular axis. Such h commutes with P, so the stabilizer is precisely the axis-ray stabilizer, of order 24. Each of the two q roots therefore gives 48,600 matrices. Their determinants are $-(11 \pm i\sqrt{135})/64$, so these two orbits are disjoint.

For the distinct-square unitary seed $\text{diag}(1,\omega,\omega^2)$, exact conjugation and group membership give stabilizer order 36 and orbit size 32,400. The two orders of ω are related by the checked generator that interchanges the last two coordinate axes. Its determinant $-1/5$ distinguishes this orbit from the other three.

The four orbits contain 130,680 nonzero F-flat matrices. An exact monomial-subgroup calculation attains diagonal intersections of 12 group matrices, 36 matrices in each nonunitary orbit and 24 in the massless orbit. The previous exhaustive diagonal census excludes further diagonal intersections. Therefore 130,572 of these certified matrices are off diagonal. Including the origin gives 130,681 certified vacua. This is a lower bound on the complete unperturbed vacuum set, not a proof that no additional or continuous components exist.

The full canonical Hessian is transported by unitary field transformations. Consequently all 97,200 nonunitary copies and all 1,080 group copies retain their exact nonsingular representative Hessians: 98,280 nonzero vacua have a certified positive eighteen-real-field mass matrix at the benchmark. The 32,400 massless copies retain complex Hessian rank six.

Massive relaxation and the massless branch

The implementation reconstructs the complete nine-variable sextic with 252 monomials directly from the stored tensor. For the normalized massless seed $K0=\text{diag}(1,\omega,\omega^2)$, use $a=-5,b=1,c=0$. Its Hessian has three complex null vectors on the off-diagonal pairs. Six complementary coordinates have an invertible Hessian Hm . Their exact inverse and every correction coefficient are retained in the receipt.

Let u denote the three null coordinates and y the six complementary massive coordinates. The holomorphic implicit-function theorem solves the six massive superpotential derivatives locally as $y=y(u)$. If $C2$ is their quadratic residual along the unrelaxed null slice, the second-order displacement is $y2=-Hm^{-1}C2$. The raw null superpotential contains a nonzero quartic polynomial. Substitution of $y2$ cancels it exactly: the effective quartic is $W4$ minus one-half $C2$ transpose $Hm^{-1}C2$, which vanishes identically.

The cubic effective superpotential also vanishes. Fifth order vanishes after the second-order displacement. At sixth order one must retain the third-order massive residual $R3$; using only $y2$ would produce a misleading nonzero answer. The exact sixth-order effective coefficient is $W6$ evaluated with $y2$ minus one-half $R3$ transpose $Hm^{-1}R3$. It too vanishes identically. Thus

$$W_{\text{eff}}(u) = W_0 + O(\|u\|^7).$$

The analytic coordinate metric remains nonsingular, so the first possible effective F-term energy is degree twelve. The result is stronger than merely finding three zero quadratic masses. It does not prove all-order flatness, a continuous moduli space, or an isolated high-order minimum. Those conclusions require higher-order or exact component analysis. The global quartic selector already excludes this entire named orbit from the global minima without relying on such an inference.

Reproduction and physical interpretation

Run ``python python/develop_valentiner_frame_selection.py`` to regenerate the scientific receipt. The focused selection, diagonal-census and canonical-SUSY suites contain seventeen tests, including independent evaluation of the full polynomial against the original tensor, generalized CP and group invariance, full original F terms, exact orbit stabilizers, all 64 tangent Gram entries, rational soft-energy inequalities and nonlinear implicit elimination through sixth order. A fresh extraction reproduces the receipt byte for byte.

The central advance is an explicit permitted soft interaction that corrects the known branch ordering, plus a calibrated scalar EFT completion with a proved global unitary tensor-frame class. The off-diagonal census and nonlinear calculation establish why a local diagonal ansatz was insufficient. This is a result about the isolated link interaction. It does not yet derive a complete quark-source frame, the golden CKM relation, physical CP selection, radiative stability of the radial calibration, or a supersymmetric gauge completion.